nanospice

A SPICE circuit simulator in 1000 lines of Rust

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github.com/milanofthe/nanospice

The rule

- one file, at most 1000 nonblank, noncomment lines, std only
- a test counts the lines and fails the build above the limit
- tests, comments, and example netlists are free

The budget forces every feature to displace another: the cost of each decision becomes explicit.

current count: **1000 of 1000**

What fits

Analyses

– `.op`, `.dc`, `.tran`, `.ac`

Devices

– R, C, L, V, I, E, G
– diode, MOSFET, JFET, BJT
– junction and gate capacitances

Solver

– sparse LU, partial pivoting
– Newton with junction limiting
– gmin and source stepping
– trapezoid, LTE, breakpoints

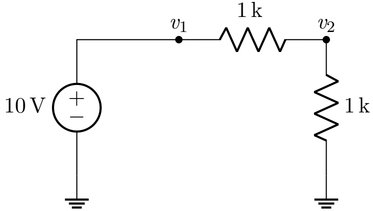
Input

– SIN, PULSE, PWL; `.model`, `.print`

Where the lines go

code section	lines	cumulative
constants and plumbing	23	23
numbers: `Num` trait, complex, suffixes	78	101
circuit data model	61	162
parser	275	437
sparse LU and stamp helpers	121	558
device models	64	622
solver	142	764
output selection	26	790
analyses and main	210	1000

Modified nodal analysis



$$\begin{pmatrix} g & -g & 1 \\ -g & 2g & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix}$$

– one branch current per voltage-defined element (V, L, E)

– the branch row has a structural zero on the diagonal: pivoting is mandatory

– ground is a dummy row 0: stamps never special-case it

Sparse LU

`sparse LU and stamp helpers` 121 lines

- rows are sorted (column, value) vectors; elimination is a merge, fill-in falls out as the union

The first-entry invariant

Every active row contains only columns $\geq k$: a column-k entry is always the *first* element, so rows bucket by their first column and pivot search reads one bucket. The factorization is linear on ladders and trees.

- a singular matrix names its column: `singular matrix at node b, no conduction path?`

Two primitives carry every device

```
fn contrib(&mut self, p: usize, n: usize, i: T,
           ctl: &[(usize, usize, T)], x: &[T]) {
    let mut ieq = i;
    for &(a, b, g) in ctl {
        self.quad(p, n, a, b, g);
        ieq = ieq - g * (x[a] - x[b]);
    }
    self.rhs(p, T::zero() - ieq);
    self.rhs(n, ieq);
}
```

The contribution operator of Verilog-A, $I(p, n) <+ f(v)$; the dual `vcontrib` writes the branch row of the voltage-defined elements.

The whole device set

device	primitive	controls
R, C, I	`contrib`	self conductance
G (VCCS)	`contrib`	one control pair
diode	`contrib`	one junction
MOSFET, JFET	`contrib`	g_m, g_{ds}
BJT	3 × `contrib`	two junctions, transport
V, L, E	`vcontrib`	none, own current, control pair

- nmos/pmos and source-drain swap in one evaluation: every derivative picks up the polarity twice, $s^2 = 1$, the stamps are sign-free
- the refactor to this form freed the budget for the JFET and PWL

Junction limiting

$$v_{crit} = V_T \ln \frac{V_T}{\sqrt{2}\, I_S} \qquad\qquad v \leftarrow v_{old} + V_T \ln\left(1 + \frac{v_{new} - v_{old}}{V_T}\right)$$

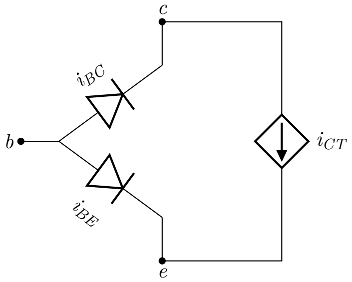
– Newton diverges on the bare exponential: 5 V proposed means e^{193} in the Jacobian

– beyond the curvature maximum v_{crit}, steps become logarithmic

Field note: limiting must deny convergence

A common-emitter amplifier converged with the transistor off: the limiter clamped the junction, no current flowed, the deltas vanished, the test said done. Repair: an iteration in which a limiter changed a voltage cannot converge. A delta test alone cannot see active limiting.

BJT: three contributions




```
s.contrib(*b, *e, .., &[(*b, *e, gpi)], x);
s.contrib(*b, *c, .., &[(*b, *c, gmu)], x);
s.contrib(*c, *e, ..,
  &[(*b, *e, gmf), (*b, *c, -gmr)], x);
```

KCL composes the classic 3×3 stamp; rows and columns sum to zero by construction.

Junction capacitances by desugaring

$$C(v) = C_{j0}\left(1 - \tfrac{v}{V_J}\right)^{-m}, \qquad Q(v) = \frac{C_{j0}\,V_J}{1-m}\left(1 - \left(1 - \tfrac{v}{V_J}\right)^{1-m}\right)$$

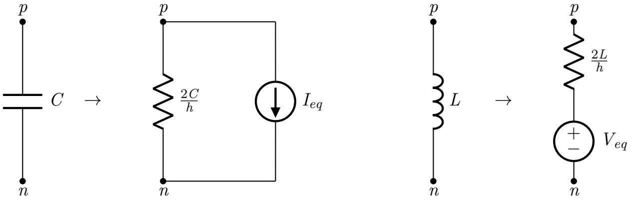
- `cjo`, `cje`/`cjc`, `cgs`/`cgd` desugar into internal capacitors in the parser
- no assembly, state, timestep, or AC code knows they exist
- charge-based, graded ($m = 0.5$) for junctions, linear for gate oxide

Operating point fallbacks are data

```
let gmin_path: Vec<(f64, f64)> =
    (2..=12).map(|k| (10f64.powi(-k), 1.0)).collect();
let src_path: Vec<(f64, f64)> =
    (1..=10).map(|k| (GMIN, k as f64 / 10.0)).collect();
for path in [gmin_path, src_path] {
    if path.iter().all(|&(g, fac)|
        newton(ckt, x, lim, &Mode::Dc { fac }, g, ITL_OP).is_some()) {
        return;
    }
}
```

Both classic fallbacks are homotopy paths in the (g_{min}, source scale) plane, walked with warm starts.

Transient: companion models



- state per reactive element: charge or flux plus the dual rate, one update rule for both
- trapezoid: second order, A-stable, undamped

Adaptive timestep

$$\varepsilon_{tr} \approx \frac{h^3/12}{h(h+d_1)(h+d_1+d_2)/6 + h^3/12}\,\big(x_c - x_p\big), \qquad h' = 0.9\,h\,r^{-1/3}$$

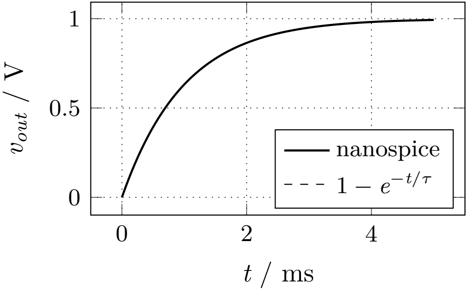
- quadratic predictor through the last three points; corrector minus predictor measures $\dddot{x}$ (uniform steps: the classic 1/13)
- the predictor doubles as the Newton starting point
- waveform corners are breakpoints; steps land on them exactly
- the two predictor-less restart steps run damped backward Euler

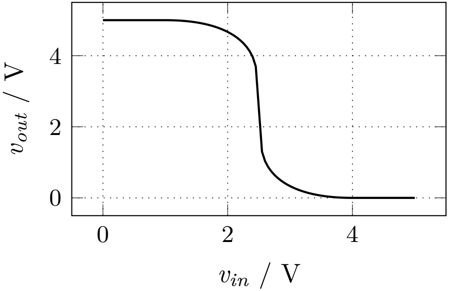
Small-signal AC

$$(G + j\omega C)\, \hat{x} = \hat{b}$$

- at a converged operating point the limiters pass voltages through: *G is the DC Jacobian the last Newton iteration assembled*
- the AC path reuses `assemble`, maps to complex, adds $j\omega C$ and $-j\omega L$
- one generic sparse LU over a three-method `Num` trait serves Newton and AC; no device model contains AC-specific code

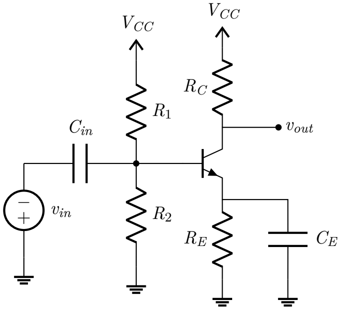
Results: step response and inverter

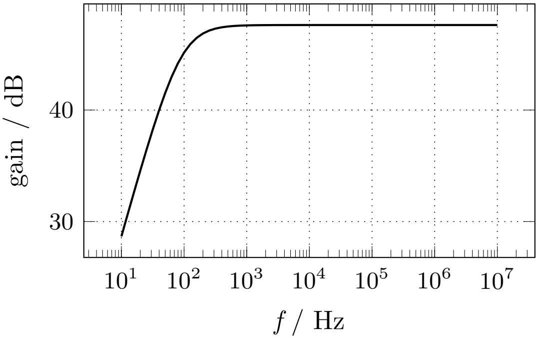




RC transient against the closed form; CMOS inverter `.dc` transfer

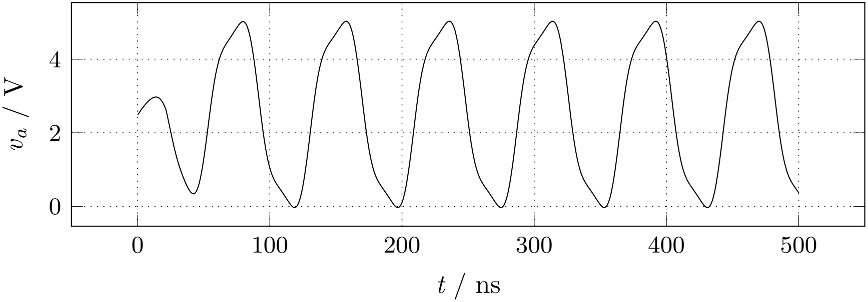
Results: common-emitter amplifier





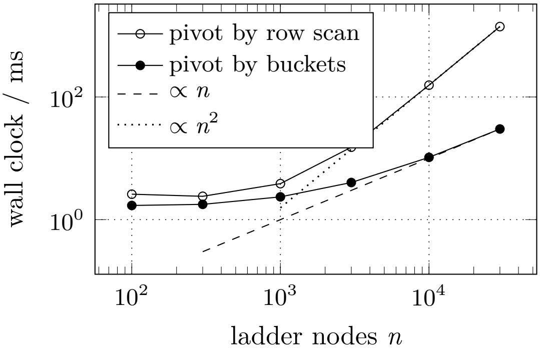
midband $g_m R_C \approx 240$, exactly 47.6 dB in the flat region

Results: it oscillates



- three CMOS inverters, gate capacitances set the stage delay
- period 78 ns, stable within 1.5 %; a test bounds the jitter
- without the capacitances the ring has no defined frequency

Benchmark: the plot that fixed the solver



- the first version scanned every remaining row per column: $O(n^2)$, exposed by this plot
- per-column buckets fixed it in eight lines
- 30000 nodes: 1404 ms $\rightarrow$ 30 ms, linear through the last point

Validation

`tests/cli.rs` — 330 lines, 0 of 1000: tests are free

23 integration tests, references independent of the implementation:

- analytic: step responses, AC corners, LC amplitude and energy
- bias points: MOSFET, JFET, BJT, exact npn/pnp symmetry
- structural: randomized ladder vs. a Thevenin reduction computed in the test
- negative: malformed decks produce named errors, never a panic

The cut list

feature	why it lost	est. lines
`.subckt` hierarchy	lost to the transistor models	$\sim$100
Markowitz ordering	netlist order suffices	$\sim$80
Gummel-Poon BJT	Ebers-Moll covers the core	$\sim$80
noise analysis	needs the adjoint system	$\sim$80
`.param` expressions	a calculator is its own project	$\sim$60

The most consequential remaining cut is `.subckt`: real decks are hierarchical.

Closing

1000 of 1000 lines

Sparse LU, two contribution primitives, Newton with limiting and homotopy fallbacks, trapezoid with LTE, breakpoints and damped restarts, graded junction capacitances, AC from the DC Jacobian, eleven devices, four analyses, 23 tests.

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MIT licensed; report with all derivations in the repository.